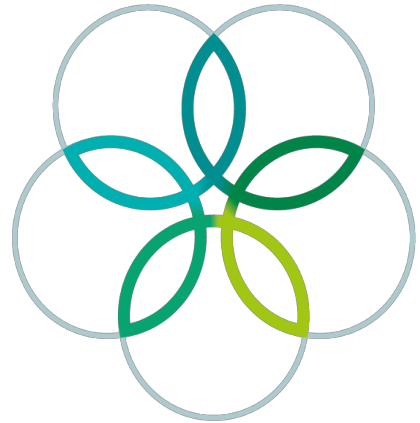
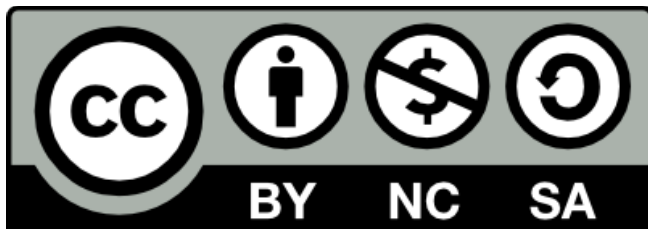


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OLYMPIAD e. V.

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PRACTICAL TEST

4th International Biology Olympiad 1993, Utrecht, the Netherlands

The test consists of five sections: I to V.

There are two groups of participants.

Group A: participants with nos. 1 - 30;

Group B: participants with nos. 31 - 60.

There are five halls, numbered 1 - 5.

You will not do all five sections of the test in the same hall. There are guides to tell you where you are supposed to go and where your seat is. You are not allowed to leave before the end of each section.

Group A will do the practical test in the following sequence:

Section I in hall 1	(60 mins.)
Section II in halls 2 and 3	(60 mins.)
Section III in hall 1	(45 mins.)
break	(15 mins.)
Section IV in hall 4	(45 mins.)
Section V in hall 5	(45 mins.)

The sequence for group B is as follows:

Section II in halls 2 and 3	(60 mins.)
Section III in hall 1	(45 mins.)
Section IV in hall 4	(45 mins.)
break	(15 mins.)
Section V in hall 5	(45 mins.)
Section I in hall 1	(60 mins.)

The maximum number of points to be obtained amounts to a total of 201.

The maximum points to be gained from sections I - V separately are as follows:

Section I:	40 points
Section II:	41 points
Section III:	45 points
Section IV:	39 points
Section V:	36 points

The maximum score for each separate question is indicated between brackets in the margin of the answer sheets.

PRACTICAL TEST
4th International Biology Olympiad 1993

Section I: The reaction of nematodes to manuring. (60 mins., hall 1)

Some 80 % of all multicellular animals on earth are nematodes. One kilogramme of soil contains an average of 30,000 individuals, representing some 30 to 60 different species.

You are going to analyze the effects of manuring on these nematodes. Preliminary research showed that a certain type of poor sandy soil contained the following types of nematodes:

plant parasites (pp)
bacteria eaters (ba)
fungus eaters (fu)
carnivores (ca) and
omnivores (om).

Dehydrated nematode-free cattle manure was added to this sandy soil. Six days later, a second sample was taken for nematode fauna analysis. We now have two samples, one taken before manuring, the other afterwards. The table below shows the composition of the nematode fauna in percentages before manuring:

Name	Food	group	%
Acrobelus	ba	2	46.5
Aphelenchoides	fu	2	4.5
Diplogaster	ba	1	1.5
Eudorylaimus	om	3	11.0
Hemicycliophora	pp	3	6.0
Prionchulus	ca	3	3.5
Rhabditis	ba	1	3.0
Rotylenchus	pp	3	9.5
Trichodorus	pp	3	2.0
Tylenchus	pp	2	5.5
Others			7.0

Each name is followed by the type of food and a number from 1 to 3. This number gives information about the life strategy of the species in question. The following characteristics are distinguished:

Group 1 Enrichment opportunists.

These nematodes can exploit new food sources fast.

They have a life cycle of several days.

They produce numerous small eggs.

They are relatively insensitive to stress.

They are transported by insects.

They are found in soil with high food density and show dormancy periods (dauerlarvae) when the food supply decreases.

Group 2 General opportunists.

These nematodes have a longer life cycle than the enrichment opportunists.

They are susceptible to stress but are not found in poor soil.

Group 3 Persisters.

These nematodes have a life-span of several months.

They produce few but large eggs.

They are relatively sensitive to stress.

The relationship between 'enrichment opportunists' (1), 'general opportunists' (2) and 'persisters' (3) is indicative of changes in the ecosystem of the soil.

The distribution among the groups in percentages:

Group 1 4.5% (3.0% Rhabditis + 1.5% Diplogaster), group 2 = 56.5% and group 3 = 32.0%. This makes a total of 93%. Allowing for the percentage of non-identified nematodes (7%), we arrive at a composition of $100/93 \times 4.5 = 4.8\%$ for group 1, 61.8% for group 2 and 34.4% for group 3.

We can construct a right-angle triangle graph on the basis of the data for group 1 (4.8%) and group 3 (34.4%), producing the point indicated in fig. I.1 (see answer sheets).

Task

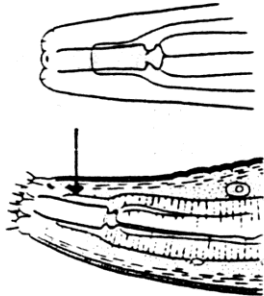
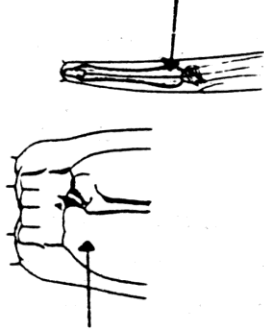
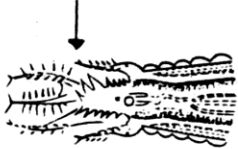
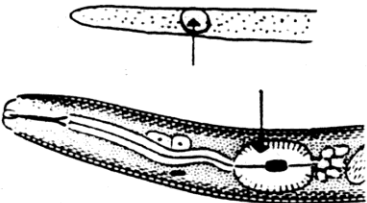
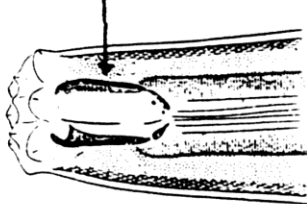
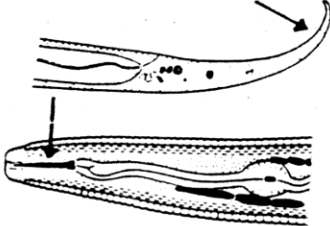
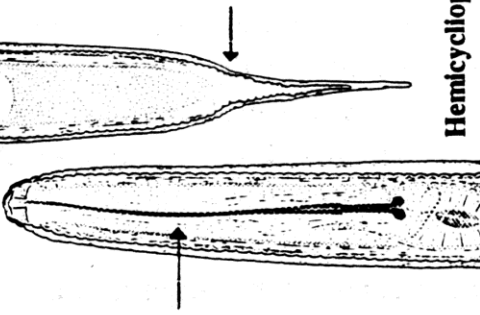
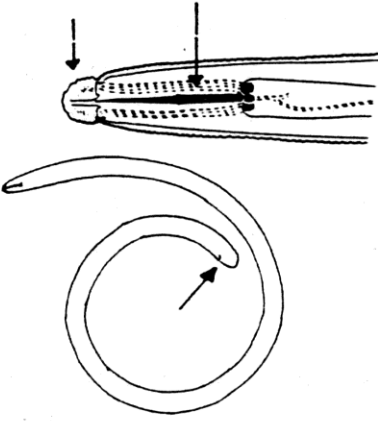
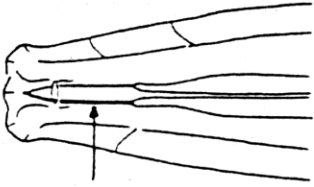
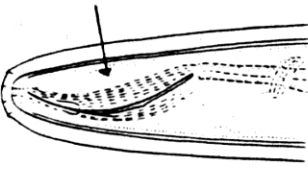
Your task is to analyze the nematode fauna (six days) after manuring. You are then required to compare your data with the situation before manuring, and interpret any changes that have occurred.

To this end, you will be given a microscopic specimen amply provided with nematodes.

- Find the nematodes with the small magnification, then use the 40x-lens and identify them on the basis of the pictures supplied.
- Return to the small magnification. Begin on the left hand side of the specimen and work through the specimen systematically and representatively until you have identified 40 nematodes. Allow at least 45 minutes for this task. Keep a record of your observations.

Process your findings as follows:

- 1 Write down your answers on the answer sheet in numbers and percentages.
- 2 Calculate the percentage ratio of the three groups of nematodes on the basis of their life strategies, and enter your answer in the answer sheet.
- 3 Then indicate these percentages in fig. I.1 on the answer sheet.
- 4 Finally, answer questions I.1 - I.7.

Answer sheets section I

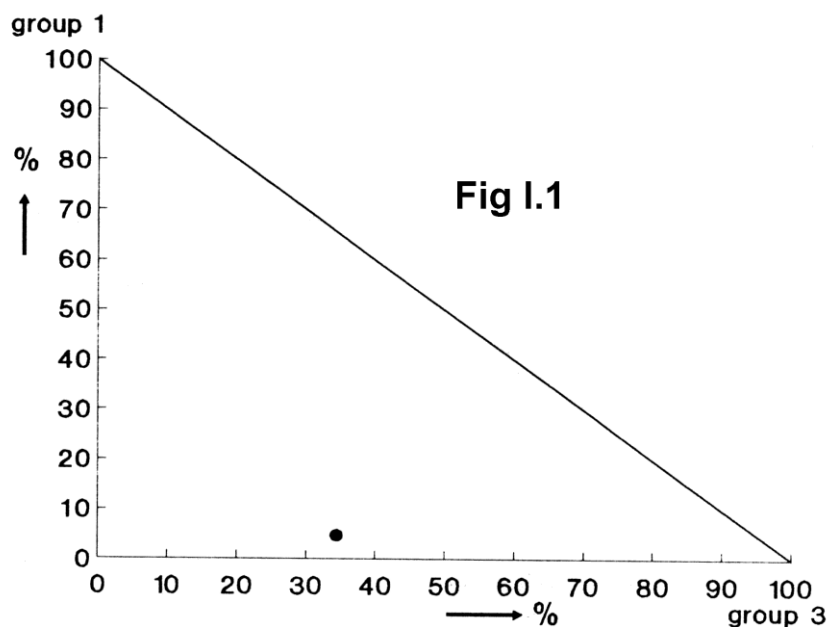
(20) 1 Results of counting and percentage calculation:

Name	Number	%
Acrobelus		
Aphelenchoides		
Diplogaster		
Eudorylaimus		
Hemicycliophora		
Prionchulus		
Rhabditis		
Rotylenchus		
Trichodorus		
Tylenchus		
Unidentified		

(3) 2 Ratio of the three groups of nematodes in %

Group	%
1	
2	
3	

(2) 3 Indicate these percentages with a point in fig. I.1.



- 4 Mark the correct answers to the following questions.
There is only one correct answer to each question.
- (2) I.1 In the triangle (fig. I.1) the point has shifted in a certain direction.
What does this indicate?
- ☐ Larger presence of predators
 - ☐ Increased food supply
 - ☐ Stress
 - ☐ Increase in number of nematodes
 - ☐ Shift within food types
- (2) I.2 Which food type has benefited most from the manure?
- ☐ Fungus eaters
 - ☐ Plant parasites
 - ☐ Carnivores
 - ☐ Bacteria eaters
 - ☐ Omnivores
- (1) I.3 The number of which species has increased most, in absolute terms?
- ☐ Prionchulus
 - ☐ Acrobeles
 - ☐ Rhabditis
 - ☐ Tylenchus
 - ☐ Rotylenchus
- (2) I.4 Looking at the adult nematodes, it is obvious that they reproduce quickly.
How can you tell?
- ☐ Many females with eggs
 - ☐ Comparatively more adult individuals
 - ☐ Darker intestines
 - ☐ Smaller females
 - ☐ Swollen gullet
- (2) I.5 After the experiment, the total number of nematodes has doubled.
Calculate the reproduction factor r (final density/initial density) for Rhabditis.
- ☐ r is smaller than 1
 - ☐ r is between 1 and 3
 - ☐ r is between 3 and 10
 - ☐ r is higher than 10
 - ☐ One of the density figures is 0
- (3) I.6 In absolute terms, the number of persisters (group 3) declines.
One possible cause of this is that
- ☐ They are affected by fungi which do not distinguish between manure and nematodes
 - ☐ They are eaten by carnivorous nematodes which have begun to develop in huge numbers
 - ☐ They are sensitive to substances like ammonia, produced in the soil.
 - ☐ The food of the persisters decreases
 - ☐ They lose the competition for space in the soil

- (3) I.7 Nematodes are fairly resistant to agitation (ploughing!) of the soil, as long as it is not too dry. If we take one kilogramme of good agricultural soil, sift it well (through a 5 mm sieve) and then leave it for a week, what can we expect?
- ☐ No changes: nematodes are quite resistant to agitation
 - ☐ Increase of group 1 due to increased microbial activity
 - ☐ Decrease of group 1 due to decreased microbial activity
 - ☐ Decrease of group 1 as a result of physical disturbance
 - ☐ After a week no persisters will be found, since they are left in the sieve.

End of section I

PRACTICAL TEST
4th International Biology Olympiad 1993

Section II: The effects of nitrogenous fertilization on the individual plant.
(60 mins., halls 2 and 3)

The plants used for this experiment, the agricultural crop maize (*Zea mays*), were grown in a greenhouse under different nitrogen conditions. These nitrogen levels are indicated with L (low) and H (high). The levels of all other nutrients were the same for both test groups. In this section, you will study the resulting growth differences.

For this purpose, you will have to

- A weigh
- B determine leaf surface
- C determine root length

You will do this in various places in halls 2 and 3.

There are not enough practical test sets for all of you. The following are available:

- 5 scales for weighing (takes very little time)
- 10 sets for determining leaf surface (takes little time)
- 20 sets for determining root length (takes a bit longer)

It makes no difference which set you use.

Plan your time in such a way that you can do all the tasks.

As you also need time to process your findings, you will hardly have to wait for each other, if at all.

Task IIA

A number of plants were harvested before this Olympiad. The leaves, stems and roots of these plants have been collected and dried separately. This is the dried material that you will find in the plastic pots and cups. Each pot contains leaves, stems or roots of three plants. The net weight of each pot is indicated on the outside.

- Determine the dry weight of the leaves, stems and roots of three plants with an (electronic) scale.
Do this for both nitrogen levels.
Write the code (LA ... or HA ...) of the pots used and the average dry weight for each plant on your answer sheet.
- Now calculate the ratio between the average dry weight of the shoot (leaves plus stem) and the average dry weight of the roots as follows:

$$\frac{\text{leaves + stem}}{\text{root}}$$

- For maximum growth under fixed nitrogen conditions, the plant could distribute its biomass across shoot and root in several ways. This would result in changes in the shoot/root weight ratio. On the answer sheet, a number of possible explanations are suggested for the differences found. Indicate in the margin with + or - which of these suggestions can and which cannot serve as possible explanations.

Task IIB

Apart from the weight of a plant, the leaf surface is also an important factor. One way of determining it is to measure the lengths of the leaves of a plant and the widths at half the length of each leaf. The average leaf surface can then be calculated as follows:

length x width x 0.761.

- The leaves of plants grown at two different nitrogen levels have been placed under transparent plates. For each nitrogen level, determine the total leaf surface of one plant.
On the answer sheet, enter both the code of the plants measured (LB ... or HB ...) and the results.

The *fresh weight* of the leaves is indicated for each individual plant under the plates. Another experiment showed that the dry weight/fresh weight ratio is equal to:

0.13 for leaves of the high-nitrogen plants

0.15 for leaves of the low-nitrogen plants.

- On the basis of these data, the calculation carried out in the first part of task IIB and the results of task IIA, calculate the average leaf surface of the high-nitrogen (H) and the low-nitrogen (L) plants you weighed for task IIA.

Task IIC

Another important factor besides leaf surface is root length. It is determined as follows. A representative piece of root is placed in a large Petri dish filled with water. The dish is placed on a grid showing parallel horizontal and vertical lines at fixed intervals. The larger the number of crossings of root and grid lines, the larger the root length. The formula for root length calculation is the following:

Root length = the number of crossings x correction factor.

The correction factor for the grids used is 0.987.

- 20 grids are available. Choose one of them.
Determine the total root length of a piece of root, for both high-nitrogen and low-nitrogen plants. On the answer sheet, enter the code of the roots (LC ..., HC ...) and your findings.

Data on the fresh weight of the roots you have weighed are supplied. The dry weight/fresh weight ratio is equal to:

0.09 for high-nitrogen roots

0.10 for low-nitrogen roots.

- On the basis of these data, the calculation carried out in the first part of task IIC and the results of task IIA, determine the average root length of the high-nitrogen (H) and the low-nitrogen (L) plants you weighed in task IIA.
- Now calculate, for both high-nitrogen and low-nitrogen levels, the ratio between average leaf surface and average root length for the plants weighed in task IIA.

You have now calculated two indexes for the plants from task IIA:

dry weight of the shoot

dry weight of the roots (task IIA), and

total leaf surface

total root length (task IIC).

- Of course, there are physiological reasons that determine the distribution of the biomass across the various organs of the plant.
Which of the indexes you have determined gives the most information on this point?
- On the answer sheet, you will find a number of arguments to justify the choice of one of the two indexes above. Indicate which of these suggestions are good arguments for the particular choice they support, and which are not.

Answer sheets section II

Task IIA

- (4) 1 Average dry weight per plant:
- | | L (code LA ...) | H (code HA ...) |
|--------|-----------------|-----------------|
| leaves | | |
| stem | | |
| roots | | |
- (2) 2 Ratio between average dry weight of shoot (stem + leaves) and average dry weight of roots:
 Low nitrogen level:
 High nitrogen level:
- (7) 3 Which of the statements below could be adequate explanations for the differences in weight ratio found?
 Indicate your views by putting + (correct explanation) or - (incorrect explanation) in the margin.
- a Under low nitrogen conditions, the roots take in too little amino acid to allow fast growth. As a result, the plant will have to invest more in root development.
 - b Low nitrogen conditions slow down the photosynthesis process. Under such conditions, a plant will have to invest more in leaf development in order to maintain a sufficient level of photosynthesis.
 - c When there is more nitrogen, the plant contains more proteins, as a result of which less of its biomass requires other carbon products. Therefore, under high nitrogen conditions the plant has to invest less in leaves and more in its roots in order to guarantee maximum growth.
 - d Under high nitrogen conditions the plant will produce more enzymes for photosynthesis, which, consequently, will increase. This means that the stomas will have to be opened further and that more water will evaporate. This water is absorbed by the roots system. As a result, the plant will have to produce more roots.
 - e Under high nitrogen conditions a few roots suffice for adequate nitrogen intake. As a result, the plant will invest primarily in leaf development.
 - f High nitrogen levels stimulate root development, since the roots are the first organs to be able to make use of the extra nitrogen.
 - g The distribution of biomass across shoot and root is determined genetically for each individual. For this reason, nitrogen conditions only influence the growth of the plant as a whole. Consequently, the differences in shoot/root ratio should be interpreted as genetic differences, and have no function.

Task IIB

- (3) 1 Total *measured* surface of the leaves.
- | Treatment | Leaves surface (cm ²) |
|-------------------|-----------------------------------|
| L (code LB) | |
| H (code HB) | |
- (5) 2 Total *calculated* average surface of the leaves from task IIA:
- | Treatment | Average leaf surface (cm ²) |
|-----------|---|
| L | |
| H | |

Task IIC

- (6) 1 Total *measured* length of representative piece of root.
- | | |
|-------------------|--------------------------------|
| Treatment | Total length of the roots (cm) |
| L (code LC) | |
| H (code HC) | |
- (4) 2 Total *calculated* average root length of the plants from task IIA:
- | | |
|-----------|--------------------------|
| Treatment | Average root length (cm) |
| L | |
| H | |
- (2) 3 *Calculated* ratio between average leaf surface and average root length of the plants from task IIA:
- | | |
|-----------|--|
| Treatment | <u>average leaf surface</u>
average root length |
| L | |
| H | |
- (2) Which index gives the most information? (mark the correct answer)
- ☐ shoot weight/root weight
- ☐ leaf surface/root length
- (6) Which of the arguments below concerning the indexes are correct, and which are not? Indicate your views by placing + (correct) or - (incorrect) in the margin.
- a The rate of photosynthesis is directly proportional to the biomass of the shoot. The rate of respiration is directly proportional to the biomass of shoot plus root. This is why the shoot weight/root weight index provides the most information.
 - b The absorption of CO₂ by the leaves and of nutriment and water by the roots are processes that take place at surfaces. For this reason, the leaf surface/root surface ratio would be the best index. The leaf surface/root length ratio is an acceptable alternative, however, and therefore the best of the two indexes given.
 - c Under certain conditions (for instance drought), leaves and roots may shrink. When they do, leaf surface or root length are no longer very helpful data. Therefore it is wiser to work with the shoot weight/root weight index.
 - d The leaf surface/root length ratio gives no information whatsoever about the investment in the biomass of the stem. As a result, this index cannot serve as a reliable indicator of growth. This does not apply to the ratio between shoot weight/root weight, which is therefore a better parameter.
 - e Leaf surface is a two-dimensional measure, whereas root length is one-dimensional. As a result, the two cannot be simply combined to calculate an index. This problem does not arise in the calculation of the shoot weight/root weight ratio, which is therefore a better parameter.
 - f Since the shoot weight and the root weight both increase under high nitrogen conditions, the leaf surface/root length ratio is a better measure.

End of section II

PRACTICAL TEST
4th International Biology Olympiad 1993

Section III: The effects of nitrogenous fertilization on the anatomy of the plant
(45 mins., hall 1)

In section II of the practical test you studied maize plants grown at two different nitrogen levels. For each of these levels, you will now make a cross-section of a piece of stalk.

The stalks have already been prepared for you for this purpose.

Dye your cross-sections with a solution of iodine in potassium iodide.

Study the anatomical structure of the stalk and the effects of nitrogenous fertilization on this structure.

- Study fig. III.1 (supplied on a separate page).
It represents a picture of part of a maize plant stem. The numbers 1 - 9 refer to the cell types, the letters a - h indicate the tissues.
You are required to name these cell types and tissues.
Use names in list III.A for this purpose.
Enter the number of your choice on the answer sheet.
Also use the cross-sections you cut yourself to complete this task.
- Plants grown at high and low nitrogen levels show anatomical differences.
Study these differences on the basis of the cross-sections you have prepared.
Try to find five striking differences between the two cross-sections.
In doing so, pay attention to cell types, tissues or parts of cell types.
Indicate the location of these differences on the answer sheet with the help of list III.A.
Then indicate the nature of the differences between high and low nitrogen conditions. For this purpose, use the letters from list III.B.
Attention: there may be several differences within a single (part of a) cell type.
Make sure that you also indicate these multiple differences on the answer sheet.
- On the answer sheet, several statements are presented as possible explanations for the differences found. Indicate which of these are correct and which are not.

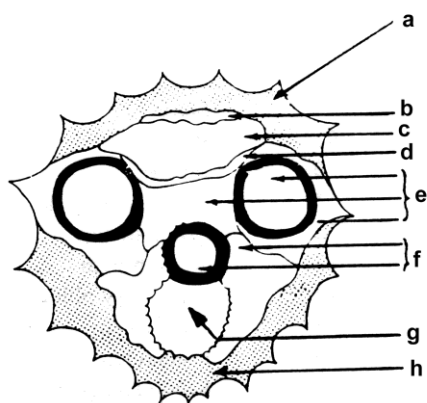
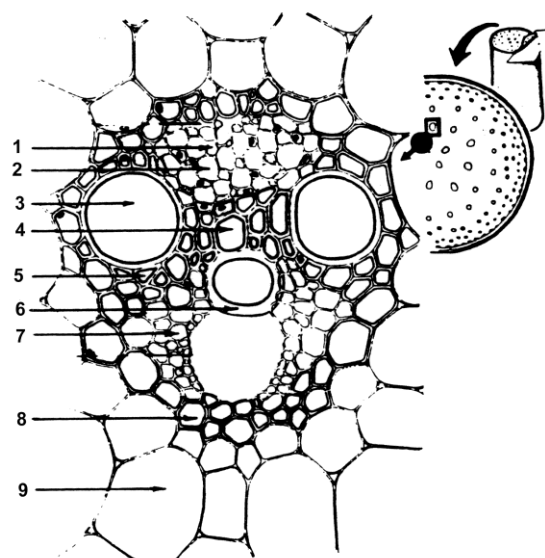


Fig III.1



LIST III.A

1. Apical Meristem
2. Lateral Meristem
3. Cambium
4. Epidermis
5. Hypodermis
6. Exodermis
7. Cortex
8. Stele
9. Pith
10. Pith Ray
11. Vascular Bundle
12. Protophloem
13. Metaphloem
14. Secondary phloem
15. Protozylem
16. Metaxylem
17. Secondary Xylem
18. Parenchyma
19. Collenchyma
20. Sclerenchyma
21. Sclerenchyma Fibre
22. Scleride
23. Vessel
24. Sieve Tube
25. Sieve Cell
26. Companion Cell

27. Subsidiary Cell
28. Xylem Fibre
29. Phloem Fibre
30. Xylem Parenchyma
31. Phloem Parenchyma
32. Tracheaid
33. Bundle Sheath
34. Pit Pair
35. Chloroplast
36. Starch
37. Diameter Cell
38. Intercellular Space
39. Primary Cell Wall
40. Secondary Cell Wall
41. Lignin

LIST III.B

- A. present
- B. absent
- C. thicker
- D. thinner
- E. larger
- F. smaller
- G. larger amount
- H. smaller amount
- I. there are more cell layers
- J. there are fewer cell layers

Answer sheet section III

(12) Cell types, tissues

Cell type (in figure III.1)	Number (in list III.A)	Tissue (in figure III.1)	Number (in list III.A)
1		a	
2		b	
3		c	
4		d	
5		e	
6		f	
7		g	
8		h	
9		i	

(25) Differences

Cell type/Tissue/Cell part (number from list III.A)	Difference at high level (letters from list III.B)

Indicate with a + or - in the margin which statements can (+) and which cannot (-) serve as possible explanations for the differences.

- (2) a Plants which are cultivated under high nitrate conditions (high nitrate plants) are able to form many new structures. In doing so, they use up photosynthesis products, preventing the build up of starch in the plants. Little energy is wasted on the storage and maintenance of reserve materials in these plants. Low nitrate plants, on the other hand, need so much energy for this that less energy is left over for growth.
- (2) b A low starch content in high nitrate plants suggests that the processing capacity is fairly equal to the photosynthesis capacity. Furthermore, it appears that the photosynthesis occurs at a faster rate in these plants than in low nitrate plants.
The slower rate of growth in low nitrate plants is the result of the smaller amount of carbohydrates available for redistribution throughout the plant.
- (2) c The slower rate of photosynthesis in low nitrate plants is a result of the fact that there are fewer proteins and fewer leaves because more roots have been formed. Because of this relatively small amount of "green" biomass, the capacity for photosynthesis is relatively small. Conversely, the larger root system leads to higher respiration. This causes the lower

- growth rate of low nitrate plants.
- (2) d Low nitrate plants, while having less capacity for photosynthesis, nevertheless have a high starch content. Therefore, this product of photosynthesis is not used for growth. It follows that not photosynthesis but another process that depends on the nitrate supply is the limiting factor for growth.

End section III

PRACTICAL TEST
4th International Biology Olympiad 1993

Section IV: Effects of nitrogen fertilization on species interaction
(45 min., room 4)

In this section we compare the reaction of two plant species to differences in nitrogen levels (nitrate and/or ammonium):

the grass *Lolium perenne* and the herb *Plantago lanceolata*.

Lolium perenne (Gramineae) is a high-growing grass (see fig. IV.1a) and is very often found in highly-fertilized agricultural grasslands.

Plantago lanceolata (Plantaginaceae) grows particularly in grasslands and roadsides poor in nutriment. It is a rosette plant; a herb without leaves on its stem (see fig. IV.1b).

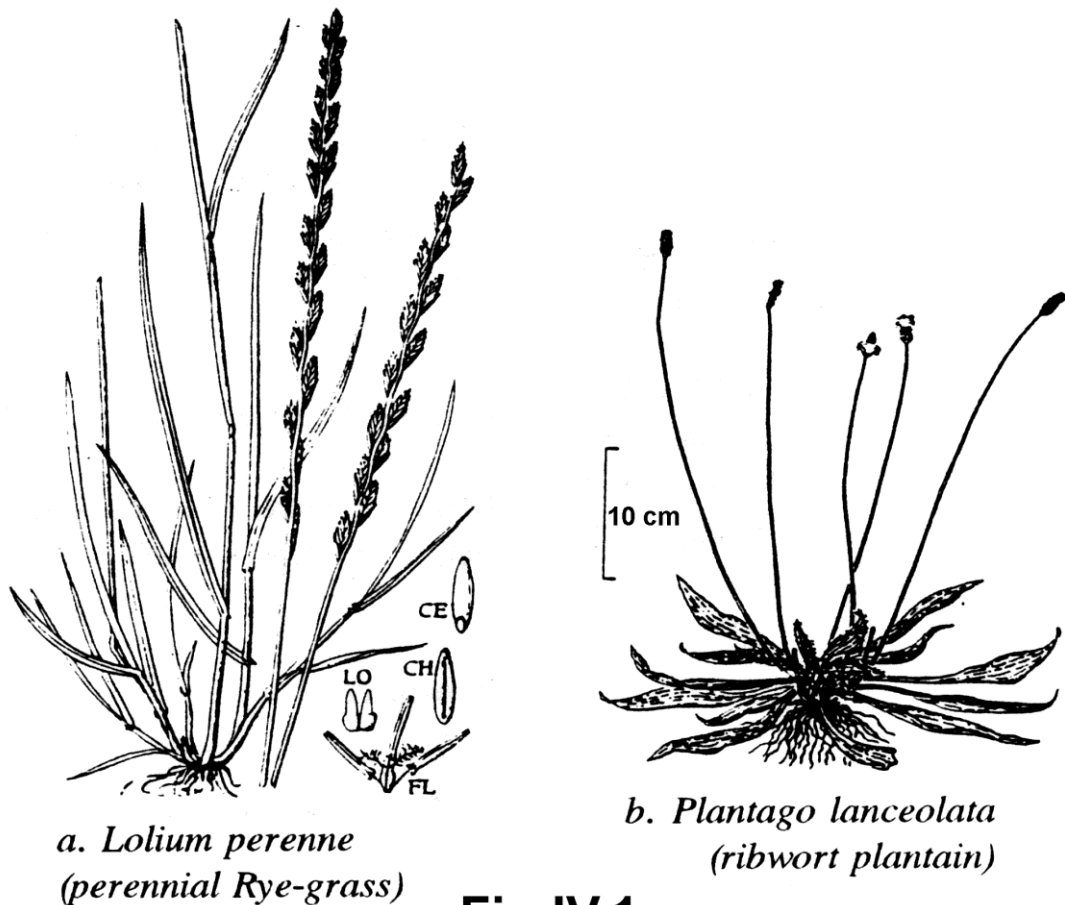


Fig IV.1

Caution: For task IV.B you must take measurements with costly light meters of which we only have a limited number. You will have to take turns. Pay attention to each other and manage your time so that you can take a turn.

Task IV.A

The increase in biomass for both species under the influence of two nitrate-fertilization levels is given in figure IV.2:

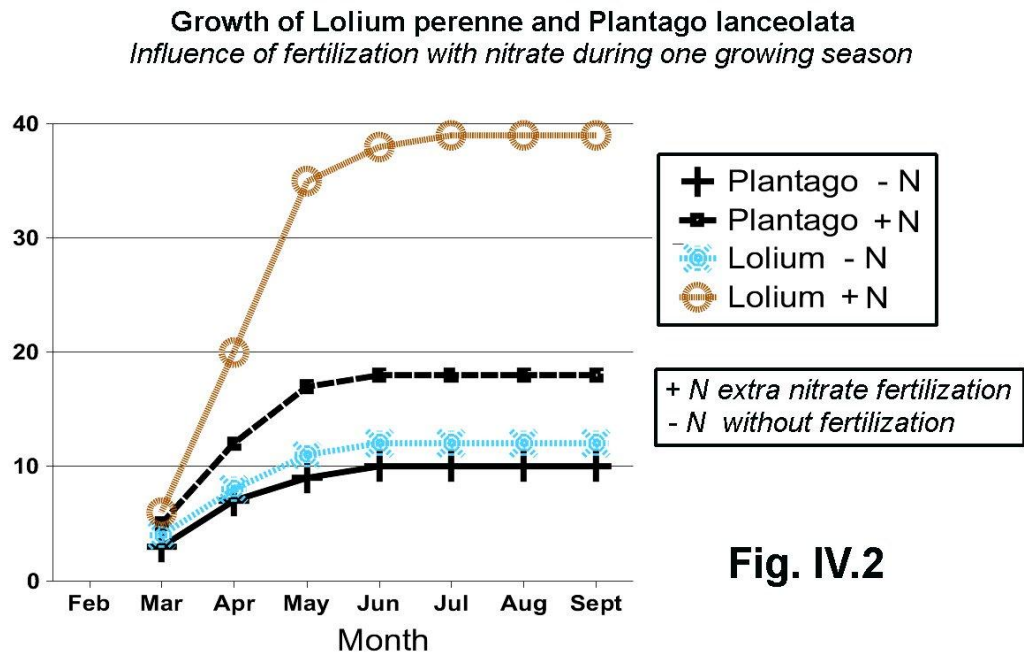


Fig. IV.2

We are concerned with the dry weight in g per pot.
The two species were grown in separate pots.
All pots are identical and rectangular: 9 cm x 10 cm.

- Calculate what the end yield is in dry weight per m² for *Lolium* as well as for *Plantago* based on the data from the graph.
Do this for the unfertilized and the fertilized case and fill in the calculated values on the answer sheet.

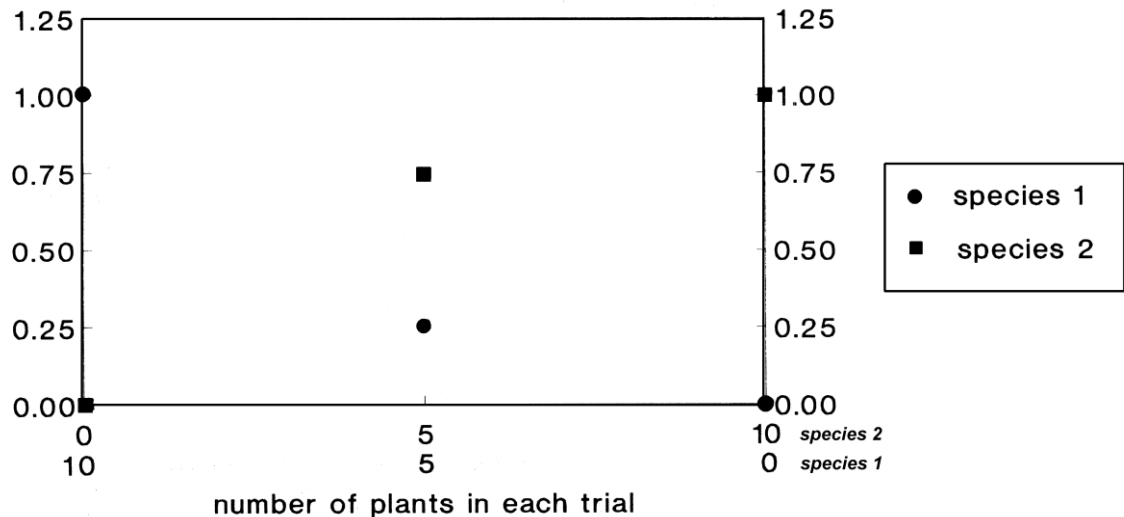
Task IV.B

In the room, there are a number with trays of vegetation from grasslands with a *high* (tray 1) and with a *low* (tray 2) nitrate-fertilization level.

- Measure the luminosity (intensity of light) at different heights with the available light meters. Measure every 10 cm starting at soil level.
Take care that the vegetation in the trays remains intact.
Make a graph on the answer sheet reflecting your findings in percentages of the original light. Choose your axes yourself.
- Rosette plants only grow moderately in the lower 10 cm of vegetation in the case of nitrogen fertilization. A number of possible explanations for this are given on the answer sheet. Indicate which suggestions can or cannot serve as explanations.

Task IV.C

The outcomes of a competition experiment between species 1 and species 2 are given in figure IV.3.



In this experiment the species were cultivated in monoculture (with 10 plants of one species per pot) as well as mixture (with 5 plants of species 1 and 5 of species 2 per pot) during a growing season. The relative yield of the mixture with regard to the monoculture is shown for species 1 and species 2 in this graph.

- The line for species 2 bends upwards and the line for species 1 bends downwards.
A number of suggestions as to why this is so are given on the answer sheet. Indicate which suggestions can and which cannot serve as possible explanations.
- Finally, indicate on the answer sheet which of the two species is a rosette plant.

End of section IV

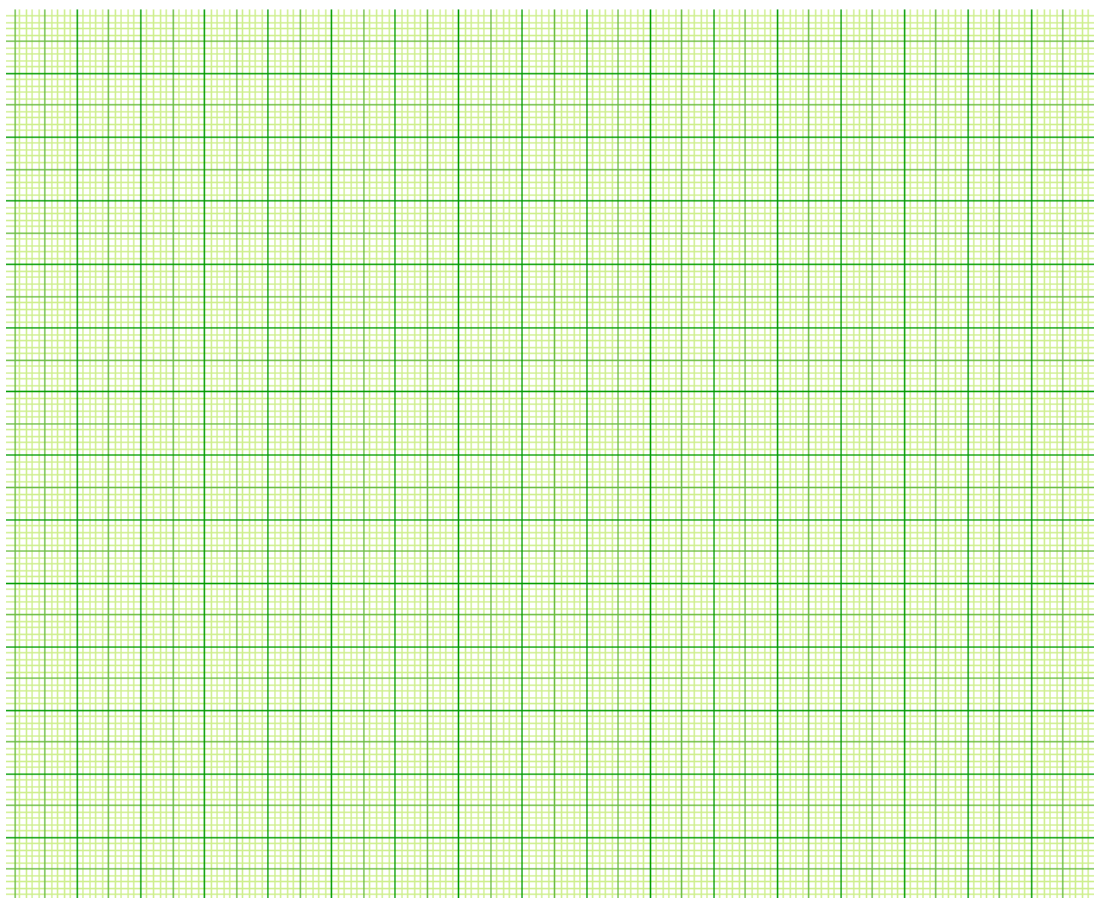
Answer sheets for section IV

Task IV.A

(3)	End yield (dry weight in g per m ²)		
	Plant	+ N: high nitrate level	- N: low nitrate level
	Lolium Perenne		
	Plantago lanceolata		

Task IV.B

(12) Graph



Task IV.B continued

- (12) Indicate with a + or - in the margin which statements can (+) and which cannot (-) serve as possible explanations.
- a Rosette plants (for example Plantago) cannot grow better under high nitrate fertilization conditions because the nitrate inhibits photosynthesis and, consequently, growth.
 - b Rosette plants are able to grow faster under low nitrate conditions because more rain water reaches the soil.
 - c The leaves of rosette plants receive little light in the case of high nitrogen fertilization and therefore are only able to grow moderately.
 - d Rosette plants develop a compact, stocky form in the case of high luminosity. Under high nitrate fertilization conditions they receive less light and therefore show higher growth and take on an ascending form.
 - e Rosette plants dry out more quickly in the case of nitrogen fertilization because all the rainwater reaches the soil by way of grass shoots, therefore benefiting the grass.
 - f The leaves of rosette plants are low in the vegetation and are overshadowed by the longer grasses after fertilization, so that they are less able to photosynthesize.
 - g The leaves of rosette plants are low in the vegetation. Little CO₂ is available there because it is used up by the higher grasses. For this reason less photosynthesis occurs.

Task IV.C

- (8) Indicate with a + or - in the margin which statements can (+) and which cannot (-) serve as possible explanations.
- a species 1 grows better in mixture than in monoculture because there is less intraspecific competition than interspecific competition
 - b species 1 grows better in monoculture than in mixture because there is less intraspecific competition than interspecific competition
 - c species 2 grows better in monoculture than in mixture because there is less intraspecific competition than interspecific competition
 - d species 2 grows better in mixture than in monoculture because there is more intraspecific competition than interspecific competition
 - e in both cases, species 1 and species 2, intraspecific competition is the most prevalent
- (4) The rosette plant is species: (fill in: 1 or 2)

End of section IV

PRACTICAL TEST
4th International Biology Olympiad 1993

Section V: Nitrogen balance of a fertilized and an unfertilized meadow
(45 min., room 5)

Task V.A

The nitrogen cycle for a West European grassland fertilized with nitrogen is given on the answer sheet.

In figure V.1, a number of processes c.q. phenomena are missing.

Complete the schematic drawing.

Name the processes (phenomena) which are *missing* from the cycle.

Choose from the provided list of processes (phenomena).

Write the corresponding numbers in the correct places in figure V.1.

Task V.B

There are a number of water samples (samples 1, 2 and 3) in the room.

One sample is collected rainwater. The other two samples are both groundwater collected at a depth of ± 50 cm. One of the samples is from an unfertilized natural grassland and the other from a nearby agricultural grassland fertilized with nitrogen.

- From each sample determine:
 - a) the acidity (pH) using the strip method.
 - b) the nitrate content using the strip method.
 - c) the ammonium content using the drip method.

The strip method is simple: dip, compare with the colouring chart and match with a colour comparison value.

The drip method for ammonium is as follows:

- 1 Rinse the measuring vessel with the water to be tested and fill to the 5 ml mark.
- 2 Add 10 drops of reagent 1 and swirl.
- 3 Add 1 microspoon of reagent 2. Dissolve by swirling and wait or set aside for 5 minutes.
- 4 Add 6 drops of reagent 3 and swirl.
- 5 After 5 minutes, place the measuring vessel on the colouring chart and match with a colour comparison value.

- Fill in your measurements on the answer sheet. Pay attention to the units of measurement.
- Fill in on the answer sheet where each sample comes from
- For both grasslands, calculate the amount of run-off ammonium + nitrate per m^2 .

Do this using the fixed concentrations in the water samples and considering the fact that the rainfall balance (precipitation minus evaporation) is 300 mm. 1 mm precipitation corresponds with 1 litre of rainwater per m^2 . Fill in your answers on the answer sheet.

Task V.C

Two photographs of grassland A and grassland B are shown in fig. V.2.



- Which grassland has the greatest variety of species?
Which grassland has been fertilized with nitrate and ammonium the most?

In many countries in Europe and America extra nitrate and ammonium is added to natural areas by "acid rain".

- On the answer sheet there are some descriptions of what could happen to the variety of plant species in originally nutrient-poor nature reserves when extra nitrate and ammonium are added by way of "acid rain". Indicate which descriptions are correct or incorrect.

Answer sheet section V

Task V.A

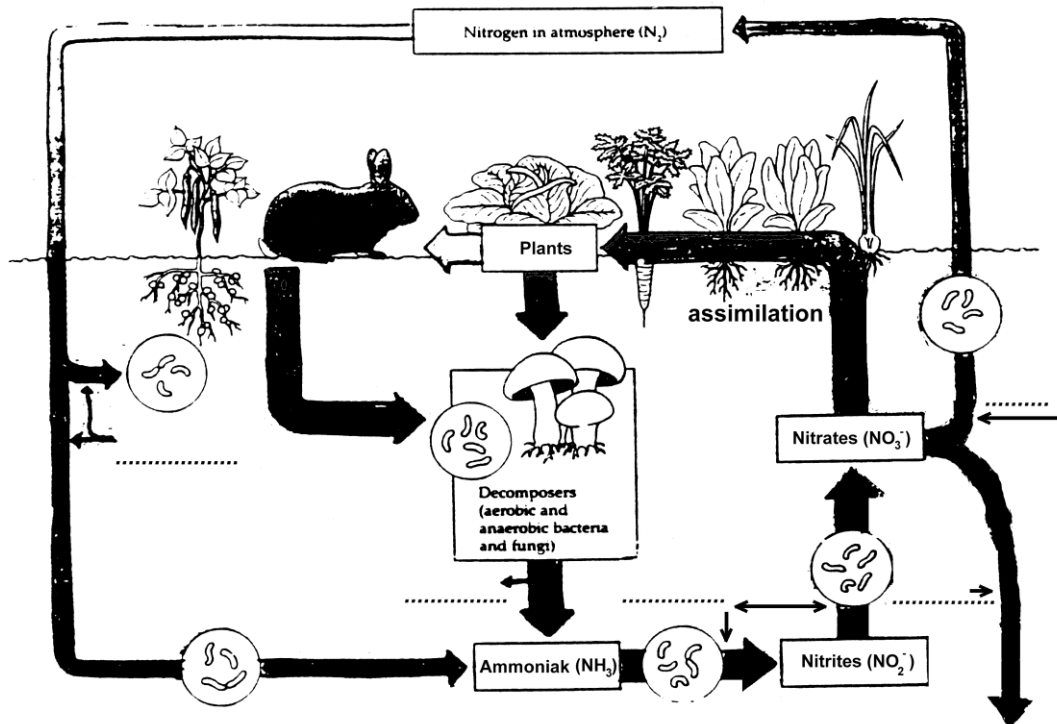


Fig. V-1

Schematic drawing of the nitrogen cycling in an agricultural grassland in Western Europe.

Complete the figure. Choose the correct processes/phenomena out of the list and write the corresponding numbers in the right places in the figure.

List of phenomena (task V.1)

- | | | | |
|-----|------------------------|-----|-------------------|
| 1) | denitrification | 12) | ammonification |
| 2) | mineralization | 13) | harvest |
| 3) | N-fixation | 14) | photosynthesis |
| 4) | acid rain | 15) | assimilation |
| 5) | nitrification | 16) | volcanic emission |
| 6) | sulphate reduction | 17) | soil weathering |
| 7) | herbivore | 18) | soil fixation |
| 8) | methanogenesis | 19) | N allocation |
| 9) | ammonia volatilization | 20) | transpiration |
| 10) | soil leaching | 21) | respiration |
| 11) | run off | 22) | retranslocation |

Task V.B

- (8) The measured values are (pay attention to units of measurement):

	pH	Ammonium	Nitrate
sample 1			
sample 2			
sample 3			

- (3) The numbers of the three water samples are:
sample number

rainwater	
fertilized grassland	
unfertilized grassland	

- (6) The amount of run-off ammonium + nitrate is:
in the unfertilized grassland:
in the fertilized grassland:

Task V.C

- (1) The grassland with the greatest variety of species is:
(fill in A or B)

- (1) The grassland that has been fertilized with nitrates and ammonium the most is:
.....
(fill in A or B)

- (10) Indicate with a + or - in the margin which descriptions can (+) and which cannot (-) serve as possible explanations.
- a the variety of plant species does not change as a result of extra nitrogen compounds in the air, because in almost all cases, phosphor limits growth and, therefore, no overshadowing by extra growth can occur
 - b the variety of plant species will decrease, because a number of fast-growing plant species (usually grasses) will increase, and the amount of light close to the ground is strongly limited causing many low-plant species to disappear
 - c the variety of species will only increase if the initial conditions were very poor in nutrients, as a result of which hardly any plant species could grow. Because of the extra nitrogen boost, more plant species can grow in the beginning and, thus, the variety of species will increase.
 - d the variety of plant species will not change, because all extra nitrogen compounds conveyed from the air, under dry as well as wet environmental conditions, are fully denitrified and are released into the atmosphere as nitrogen gas
 - e the variety of species will not change because the soil has the effect of a buffer, so that the concentration of nitrogen compounds available to the plant is kept practically constant